

# AIVIS Enterprise Platform

## AI Vehicle Insurance Investigation System — Master Technical Specification

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| <b>Target Audience:</b> Executives, Engineers, Investigators, Auditors | <b>Status:</b> Production Ready                |

### 1. Executive Summary & Application Overview

**AIVIS (AI Vehicle Insurance Investigation System)** is a multi-tenant enterprise SaaS platform designed to detect, investigate, and prevent complex automotive insurance fraud. Traditional manual investigation approaches fail against organized crash syndicates, odometer rollback, ECU tampering, and staged total-loss claims. AIVIS unifies high-density vehicle digital forensics, multi-agent AI debates, Neo4j knowledge graph analytics, and OEM telematics gateways into a court-admissible forensic workspace.

### 2. Full-Stack Technology Architecture

| Layer          | Technology / Framework                     | Key Capabilities                                                          |
|----------------|--------------------------------------------|---------------------------------------------------------------------------|
| Frontend       | React 19 + TypeScript + Vite + TailwindCSS | White Claymorphism UI, Framer Motion animations, Recharts, Lucide Icons   |
| Backend        | FastAPI (Python 3.11) + Uvicorn ASGI       | 45 REST API routes under /api/v1, OpenAPI docs, async handlers            |
| Relational DB  | PostgreSQL 15 + SQLAlchemy 2.0 ORM         | Normalized claims, vehicles, companies, users, and audit logs             |
| Cache & Queue  | Redis 7 + Celery Task Queue                | Session tokens, sliding rate limits (10k RPM), async OCR/Vision jobs      |
| Graph Database | Neo4j 5 Graph DB + Cypher                  | 15 entity node types, PageRank centrality, Louvain community clusters     |
| DevOps         | Docker Compose + Kubernetes HPA            | Multi-stage Docker builds, auto-scaling, production environment readiness |

## 3. Complete Module Breakdown (Phases 1 to 4)

### Phase 1 — Core Enterprise SaaS Infrastructure

- **SOC Command Center Dashboard (/dashboard):** Executive KPIs, active claim dossiers (89 Active), fraud risk breakdown, weekly claims vs fraud trend charts.
- **Authentication & RBAC:** OAuth2 JWT tokens with sliding refresh sessions and 5 RBAC roles (SUPER\_ADMIN, INSURANCE\_ADMIN, LEAD\_INVESTIGATOR, FIELD\_INSPECTOR, AUDITOR).
- **Claims & Vehicle Registry (/claims, /vehicles):** Full claim dossier lifecycle management, vehicle history, odometer records, VIN decoding.
- **Insurance Tenants & Audit Trail (/companies, /activity-logs):** Multi-tenant organization scoping and immutable SOC activity logger.

### Phase 2 — Vehicle Digital Forensics, Sensor Intelligence & AI Engine

- **OBD-II Acquisition Engine (/forensics/obd):** USB/Bluetooth/Wi-Fi adapters, CSV/JSON/XML importer, VIN/ECU identity, DTC codes (Active, Pending, Permanent), PID sensor stream.
- **Sensor Intelligence Engine (/forensics/sensors):** 7 sensor categories (Engine, Electrical, Transmission, Chassis, Safety, Environmental, TPMS) calculating Anomaly Scores & Multi-sensor Correlation rules.
- **CAN Bus Cyber Forensics (/forensics/canbus):** ASC/BLF/PCAP parser, 11-bit/29-bit CAN frame decoder, Attack Vector Detector (Replay, Signal Injection, Bus Flooding, Message Spoofing).
- **ECU & EDR Crash Reconstruction (/forensics/ecu, /edr):** Firmware audit, SHA-256 calibration hash, CDR crash log analyzer with 5s pre-impact speed/steering/G-force stream.
- **Digital Evidence Locker (/forensics/evidence):** Immediate SHA-256, SHA-512, MD5 signatures, EXIF camera metadata, GPS tags, immutable chain of custody trail.
- **OCR & AI Damage Vision (/forensics/ocr, /damage):** Invoice/license OCR text extraction with font forgery flags; Computer vision bounding box damage severity & cost auditor.

### Phase 3A — Enterprise Fraud Intelligence & Knowledge Graph Analytics

- **Knowledge Graph Explorer (/intelligence/graph):** Neo4j SVG canvas mapping 15 entity node types (Vehicle, Owner, Driver, Policy, Claim, Workshop, Surveyor, Bank Account, Phone, Email, IP, Device, Document).
- **Graph Mining Algorithms (/intelligence/algorithms):** PageRank Centrality rankings, Louvain Community Modularity clusters, Shortest Path tracing between suspect policyholders and body shops.
- **VIN Cloning & Syndicate Dossiers (/intelligence/vin-policy, /entities):** Cross-carrier registration audit, duplicate policy pings, shared identity fingerprinting, named syndicate ring dossiers (\$1.48M USD fraud value).

### Phase 3B — Ask AIVIS & Multi-Agent Investigation Engine

- **Ask AIVIS Assistant (AskAIVISDrawer.tsx):** Microsoft Security Copilot style conversational chat drawer with streaming Markdown, 14 quick prompt chips, pin message, copy, clear, and export.
- **11 Specialized AI Forensic Experts (/forensics/copilot):** Chief Investigator, OBD Expert, CAN Bus Expert, Vehicle Dynamics Expert, Crash Reconstruction Expert, Insurance Compliance Expert, Fraud Analyst, Digital Evidence Examiner, Document Examiner, Computer Vision Expert, Legal Advisor.
- **Multi-Agent Conflict Resolution & XAI (/forensics/explainable-ai):** Automated debate engine resolving dissenting opinions; SHAP Waterfall & LIME feature risk attribution plots.

### Phase 3C — Enterprise Operations & Compliance Suite

- **Workflow & SLA Engine (/operations/workflow, /sla):** KanBan task board, forensic checklist enforcement, real-time SLA countdown timers with Chief Risk Officer (CRO) escalations.
- **Digital Signatures PKI & Compliance (/operations/approvals, /compliance):** RSA-2048 & ECDSA-P256 PKI digital signatures; ISO 27001, ISO 21434, NIST CSF, ISO 27037 compliance audit scores & PDF exporter.

### Phase 4 — Commercial SaaS Platform & OEM Integrations

- **OEM Telematics Gateways (/commercial/oem):** Tesla Fleet API, GM OnStar (AACN), FordPass Connect, BMW ConnectedDrive, Geotab/Samsara, Guidewire ClaimCenter core bus.
- **Predictive AI & Developer Portal (/commercial/predictive, /developer):** ML models for Fraud % (94.2%), Component Failure Risk (88.5%), Cost Inflation (+276.3%); API key manager (aivis\_live\_sk\_...).
- **SaaS Licensing & Executive Portfolio (/commercial/licensing, /portfolio):** Multi-tenant quota meters, active seat counts, total insured vehicles (142,500), active claims value (\$14.85M USD), AI F1-Score (97.3%).

## 4. Key Technical Novelties & Innovations

| Novelty # | Innovation Name                    | Technical Significance                                                                                                           |
|-----------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Novelty 1 | Multi-Agent AI Debate Engine       | 11 specialized AI agent personas debating contradictory telemetry in parallel to synthesize weighted, court-admissible verdicts. |
| Novelty 2 | Neo4j Knowledge Graph Mining       | 15 entity node types executing PageRank and Louvain Modularity algorithms to uncover hidden organized fraud rings.               |
| Novelty 3 | CAN Bus Cyber Attack Detector      | Raw binary CAN log parsing (.asc, .blf, .pcap) detecting Replay Attacks, Signal Injection, Bus Flooding, and Spoofed ECUs.       |
| Novelty 4 | ISO 27037 Cryptographic PKI Locker | Immediate SHA-256/SHA-512 evidence vault hashing paired with RSA-2048/ECDSA-P256 digital signature chains.                       |
| Novelty 5 | Mathematical Explainable AI (XAI)  | SHAP Waterfall plots and LIME feature attribution providing transparent mathematical justification for AI fraud scores.          |
| Novelty 6 | Universal OEM Gateway Bus          | Unified telemetry ingestion layer standardizing streams from Tesla, GM OnStar, FordPass, BMW, Geotab, and Guidewire.             |

## 5. Complete OpenAPI REST API Specification (45 Routes)

| Module     | Method | Endpoint Path                     | Description                                     |
|------------|--------|-----------------------------------|-------------------------------------------------|
| Auth       | POST   | /api/v1/auth/login                | JWT token generation & sliding sessions         |
| Claims     | GET    | /api/v1/claims                    | List claim dossiers with status & risk filters  |
| Claims     | POST   | /api/v1/claims/{id}/verdict       | Submit forensic verdict & claim decision        |
| Forensics  | GET    | /api/v1/forensics/obd/session     | OBD-II acquisition PIDs & DTC audit             |
| Forensics  | GET    | /api/v1/forensics/canbus/frames   | Decode CAN Bus frames & attack vectors          |
| Forensics  | GET    | /api/v1/forensics/ecu/profile     | ECU firmware & SHA-256 calibration hash         |
| Forensics  | GET    | /api/v1/forensics/edr/record      | EDR crash 5s pre-impact speed & G-force stream  |
| Forensics  | GET    | /api/v1/forensics/evidence/locker | Cryptographic evidence locker & EXIF audit      |
| Graph      | GET    | /api/v1/intelligence/graph        | Neo4j Knowledge Graph nodes & edges topology    |
| Graph      | GET    | /api/v1/intelligence/algorithms   | PageRank centrality & Louvain community rings   |
| Copilot    | POST   | /api/v1/copilot/chat              | Ask AIVIS natural language query execution      |
| Copilot    | GET    | /api/v1/copilot/agents/debate     | 11 Specialized AI Expert debate findings        |
| Operations | GET    | /api/v1/operations/compliance     | ISO 27001 / ISO 21434 compliance scores & certs |
| Commercial | GET    | /api/v1/commercial/oem            | OEM telematics gateway status (Tesla, GM, Ford) |
| Commercial | GET    | /api/v1/commercial/predictive     | Predictive ML models (Fraud %, Failure risk)    |

## 6. Production Readiness & Build Verification

- **TypeScript & Vite Build:** Built cleanly with npm run build transforming 2,680 modules in 8.09s with **zero errors**.
- **Backend API Endpoints:** 45 REST API routes registered under FastAPI main.py with Pydantic v2 validation.
- **Docker Cloud Deployment:** Configured with docker-compose.yml running Frontend (Nginx), Backend (FastAPI), PostgreSQL 15, Redis 7, and Neo4j 5.